

Deduction of the Black-formula using Margrabe

The time-0 price of an exchange option is given by Margrabe's formula:

$$C_0 = S_0^1 N(d_1^M) - S_0^2 N(d_2^M),$$

with

$$d_1^M = \frac{\ln(S_0^1/S_0^2) + \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}}, \quad d_2^M = d_1^M - \Sigma\sqrt{T},$$

where $\Sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$.

Now consider a European call on the forward with payoff $(F_T - K)^+$ at time T . We now rewrite the payoff as an exchange option. Define the two assets $S_T^1 := F_T$ and $S_T^2 := K$, then

$$(F_T - K)^+ = (S_T^1 - S_T^2)^+.$$

The time-0 prices of these assets are

$$S_0^1 = P(0, T) F_0, \quad S_0^2 = P(0, T) K.$$

The first asset $S_T^1 = F_t$ has volatility σ . The second asset $S_T^2 = K$ is a constant (deterministic), so its volatility is $\sigma_2 = 0$. Hence the Margrabe volatility reduces to

$$\Sigma = \sqrt{\sigma^2 + 0 - 0} = \sigma.$$

Applying Margrabe's formula with $S_0^1 = P(0, T)F_0$ and $S_0^2 = P(0, T)K$:

$$\begin{aligned} C_0 &= S_0^1 N(d_1^M) - S_0^2 N(d_2^M) \\ &= P(0, T)F_0 N(d_1^M) - P(0, T)K N(d_2^M) \\ &= P(0, T) (F_0 N(d_1) - K N(d_2)) \end{aligned}$$

with

$$\begin{aligned} d_1^M &= \frac{\ln\left(\frac{P(0, T)F_0}{P(0, T)K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \\ &= \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \\ &= d_1, \\ d_2^M &= d_1^M - \sigma\sqrt{T} \\ &= d_1 - \sigma\sqrt{T} \\ &= d_2. \end{aligned}$$

This is exactly Black's formula.